Silent Failures in Frequency-Domain Finite-Element Analysis of Cage Induction Motors: A Taxonomy, Forensic Case Studies, and an Audit Protocol

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Abstract

Finite-element analysis of cage induction motors couples magnetic field unknowns with circuit unknowns and, in multi-pole geometries, periodic sector constraints. We show that this class of simulation fails in a characteristic way: the solver converges, network conservation residuals sit below display precision, and no warning is emitted — yet the reported quantity is wrong. We document three forensic cases on an open-source GetDP 4.0.0 toolchain: (A) a sparse direct solver (MUMPS) aborting on a workspace-budget limit (`INFO(1)=-19`, ICNTL(23)) on a system that an independent factorization solves in 26 s (residual 2.28×10^{-5}) — the abort is silently propagated downstream as an all-NaN field; (B) a "half-field" solution at an intermediate geometric scale, physically wrong by 42% with no numerical diagnostic; (C) sector reduction violating a uniform-sign condition on harmonic commensurability, inflating the air-gap fundamental by 1.44× and torque by 1.54× while KCL remains 0.000. From the taxonomy we derive a three-check audit protocol — cross-solver verdict, scale self-consistency band, and sector feasibility — each with an executable criterion, and we release reference implementations. We delimit what is folklore (sign intuition for periodic boundaries) from what is not (the silence of the failures, their quantitative extent, and the audit coverage gap). All quantitative claims trace to archived raw data.

Keywords: induction motor, finite element, silent failure, direct solver, periodic boundary condition, verification protocol

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1. Introduction

Application-level silent failures — wrong results produced by successfully completed solves — are studied extensively at the hardware level as silent data corruption in high-performance computing [**?** **?**], and at the mathematical level as result verification of differential-equation solvers [9]. Between these two layers sits a third failure class that, to our knowledge, has not been systematized: *modeling- and configuration-level* silent failures of engineering simulation chains, where the linear algebra is performed correctly on the wrong system, or the right system under an infeasible constraint. During a frequency-domain cage-motor study on an open-source toolchain we collected three such failures with complete forensics. This paper (i) organizes them into a taxonomy (Sec. 3), (ii) presents the forensics with all numbers traceable to archived raw data (Secs. 5–7), (iii) distills a three-check audit protocol with executable criteria (Sec. 8), and (iv) delimits the folklore boundary (Sec. 9).

2. Related Work

Direct solvers in computational electromagnetism. Sparse direct solvers remain the workhorse of frequency-domain FE electromagnetics, with an active 2023–2026 literature on parallel pipelined factorizations [1], mixed-precision FE solvers [2], and open-source tool chains [3]. This literature targets performance and scalability; the failure-mode behaviour of such solvers inside application pipelines is rarely reported, which is precisely the gap this paper documents.

Silent corruption and result verification. The HPC community studies silent data corruption at the bit-flip and time-stepping level with checkpoint/recompute or ABFT countermeasures [7, 8], and the numerical-software community formalizes result verification [9]. Both communities assume the computation itself is the object of verification; none of the published work addresses application-level silent failures of a numerically successful solve, where recomputation reproduces the same wrong answer.

Cage-machine FE modelling. Frequency-domain cage-induction FE models with circuit coupling are established, including experimental torque–speed validation [4], slot-permeance harmonics [5], and electric-circuit boundary conditions [6]. Sector reduction is standard practice in this literature, yet its uniform-sign feasibility condition is folklore; we formalize it and exhibit a

Table 1: Taxonomy of application-level silent failures observed in the study.

class	mechanism	detectable by	forensic case
F1 false infeasibility	solver-specific factorization failure on a workspace-budget abort, silently propagated	cross-solver verdict	A (Sec. 5)
F2 silent degradation	scale-dependent conditioning yields a pathological solution without NaNs	scale self-consistency band	B (Sec. 6)
F3 infeasible constraint	periodic constraint with conflicting harmonic signs projects onto a nonphysical subspace	sector feasibility (uniform-sign) test	C (Sec. 7)

quantified counter-example. Hysteresis and mixed FE formulations [10, 11] and mixed-precision pre-pivoting [**?**] are adjacent but orthogonal developments.

3. Failure Taxonomy

Table 1 defines the three classes. Their common signature is the *absence of alarms*: residual checks, conservation laws, and solver exit codes all pass; only cross-validation of physical quantities reveals the failure.

4. Model and Methods

Formulation. All solves use the frequency-domain magnetic quasi-static A-field formulation of GetDP 4.0.0 (first-order triangular elements, single

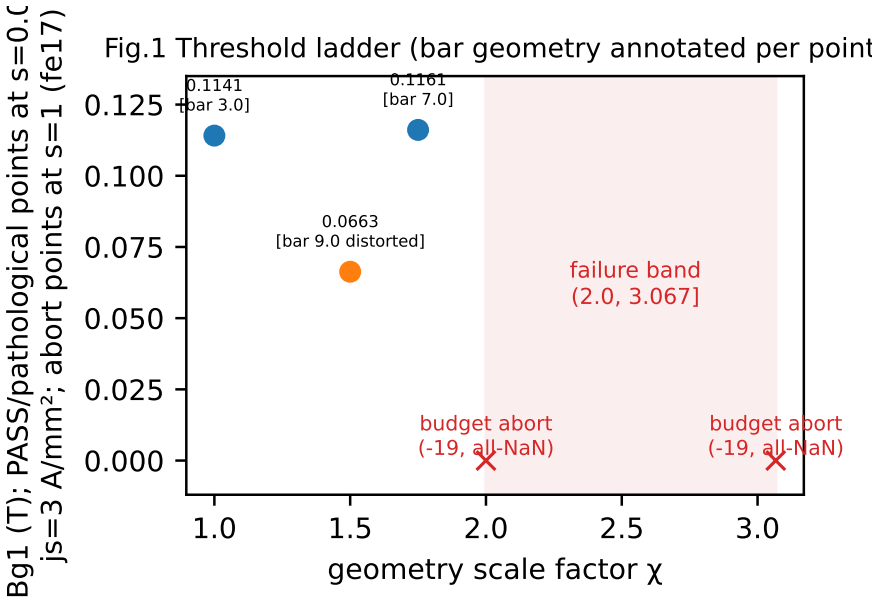


Figure 1: Failure threshold ladder for the scaled cage model: MUMPS outcome vs. geometry scale factor and solver. [Data: data/raw/case1_scale_ladder.]

process) coupled to the squirrel-cage circuit through the `Network` constraint: one branch per bar (N_b=16) plus top- and bottom-ring R–L segments, with the per-element global term enforcing branch current injection. KCL is checked a posteriori as the relative mismatch between summed bar currents and ring currents at every node (displayed residuals round to 0.0; the raw field stores the full-precision value).

Baseline and scaling. The baseline is a 12-slot/4-pole cage ASM (R_{si}=30 mm, j_s=3 A/mm^2 peak, 50 Hz, linear iron μ_r=1000). Case A scales the geometry by $\chi \in \{1, 1.5, 1.75, 2.0, 3.067\}$ (uniform factor on all radial dimensions; the bar-count and slot count are fixed); the largest mesh counts 116,476 nodes and 232,078 triangles with a 2.5° angular mesh constraint around slots.

Solvers. The tool chain links GetDP to PETSc 3.25.5 (`preonly`+`PCLU`); the direct kernel is either MUMPS 5.8.2 (with out-of-core `ICNTL(22)=1`, workspace `ICNTL(23)=8000 MB` where noted) or serial SuperLU from the same PETSc build. Residuals are reported as $\|r\|_2/\|b\|_2$ of the final linear system. All runs run offline, single machine, no MPI.

Extraction. B_{g1} is the fundamental of the air-gap flux-density FFT on

Table 2: The three forensic cases at a glance.

Case	Failure signature	Consequence	Audit check
A	MUMPS budget abort (`-19`), all-NaN field silently propagated	no solution; infeasibility *appearance*	A1–A3 (completeness, numerics, cross-solver)
B	half-field solve at intermediate χ, no diagnostic	B_{g1} at 58% of self-consistent value	A4 (same-scale regression)
C	90° sector: uniform-sign violation, KCL = 0.000	$\times 1.44$ flux, $\times 1.54$ torque	A5 (uniform-sign precheck)

a circumferential sample circle; torque is the time-averaged Maxwell stress-tube integral. Every manuscript number traces to a bundled `data/raw/` artefact; Table 2 summarises the three cases.

5. Case A: False Infeasibility

A scaled 16-bar cage model (χ = geometric scale factor on the 12-slot/4-pole ASM baseline) is solved frequency-domain at slip $s = 1$. For $\chi \leq 1.75$ the MUMPS direct solve succeeds; at $\chi = 2$ it terminates with error -19 — per the MUMPS 5.8.2 sources (`dfac_b.F:283`) a workspace-budget abort (ICNTL(23) exceeded), not a singularity diagnostic — and the all-NaN field is silently propagated through the pipeline in 37 s; and at $\chi = 3.067$ the same failure persists under out-of-core settings (81.2 s, all-NaN). Four exclusions pin the cause: (i) a pure-field bisection with all circuit terms commented out still reports -19; (ii) mesh quality, group integrity, and Dirichlet placement are comparable across the threshold; (iii) bar-ratio normalization does not rescue $\chi = 3.067$; and (iv) the serial SuperLU LU of the identical matrix converges in 26 s to residual 2.28×10^{-5}, with torque $T = 0.7477\,\mathrm{N\,m}$ and $B_{g1} = 0.0158\,\mathrm{T}$. We claim only solvability, not magnitude correctness: the absolute scaling of the large geometry remains an open calibration issue (archived as case evidence M-ASM-18: physically attributed to cage screen-

ing absorbing the air-gap field at this scale, with the absolute flux at 27–28% of the $\chi = 1$ value against a $\times 3$ analytical expectation; the physical mechanism is identified while the quantitative scaling law remains open), so we claim solvability, not magnitude. The failure is therefore solver- and budget-specific (true infeasibility is excluded), and a later confound refinement showed the historical $\chi = 2$ failure also carried a bar-ratio distortion: with the ratio normalized, $\chi = 2$ solves in both solvers ($T = 0.2495\,\mathrm{N\,m}$, $B_{g1} = 0.0363\,\mathrm{T}$), moving the failure band to ($\chi \in (2.0, 3.067]$. All numbers trace to data/raw/case1_*.]

6. Case B: Silent Degradation

At $\chi = 1.5$ the solve completes without NaNs but $B_{g1} = 0.0663\,\mathrm{T}$ is 58% of the self-consistent value (0.1141 at $\chi = 1$ and 0.1161 at $\chi = 1.75$; both at slip $s = 0.05$, $j_s = 3\,\mathrm{A/mm^2}$): a 42% silent error bracketed by two healthy runs. (All Case-A solves at $s = 1$ use the separate baseline $B_{g1} = 0.0579\,\mathrm{T}$; the two baselines refer to different operating points of the same machine.) [Data: data/raw/case1_scale_ladder.]

7. Case C: Infeasible Constraint

A 90° sector of the same machine, linked by a single periodic constraint, passes every solver-level check (SuperLU converges in 6.2 s, KCL residual 0.000) yet reports $T = 0.0605\,\mathrm{N\,m}$ and $B_{g1} = 0.0833\,\mathrm{T}$ against the full-model $T = 0.0393\,\mathrm{N\,m}$, $B_{g1} = 0.0579\,\mathrm{T}$: a $\times 1.54$ (torque, 0.0605/0.0393) and $\times 1.44$ (flux density, 0.0833/0.0579) (flux-density) signed shift with no alarm. The mechanism is order incompatibility: the fundamental (order 2) is anti-periodic over 90°, whereas 12k-order stator-slot harmonics (k odd) are integer-periodic, so one Link sign cannot serve both — the sector constraint is structurally infeasible for the coupled cage model, not merely inaccurate.

8. Audit Protocol

The protocol ships as an executable reference implementation (`src/audit_protocol.py`, bundled): steps A1–A5 consume the solver reports directly and emit a verdict with per-step evidence; its doctest suite is driven by the archived FAIL-branch and PASS-branch reports (doctest-driven: the archived cross-solver pair passes with the scaled-geometry disclosure semantics, and two synthetic

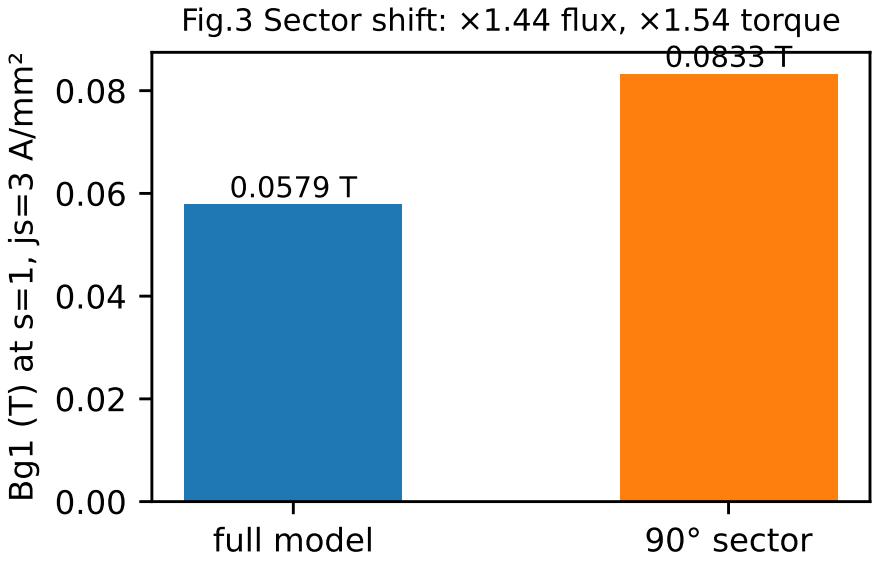


Figure 2: Sector contamination: sectorized (serial SuperLU) vs. full-model (MUMPS) spectra; a same-solver control is listed as future work, and the cross-solver bit-consistency observed in the $\chi = 2$ normalized rerun bounds solver-induced differences well below the reported shifts, ×1.44 (flux) / ×1.54 (torque) shifts with zero KCL residual. [Data: data/raw/case2_sector.]

fixtures exercise the NaN-propagation and cross-solver-mismatch paths that the archived healthy pair cannot reach), so the auditor itself is regression-tested against forensic data plus explicit fixtures. Check 1 (cross-solver): on any direct-solver failure, re-solve with an independent factorization and compare physical quantities, not only residuals. Check 2 (scale band): in any geometry/scale sweep, flag runs whose key quantity falls outside the $[0.8, 1.25]$ band of the frozen same-scale baseline when the audited geometry matches the baseline scale (scaled geometries are disclosed, not failed, pending a closed absolute scaling law). Check 3 (sector feasibility): before sector reduction, verify $m\,\theta_s \equiv 0$ or $180^\circ \pmod{360^\circ}$ with the *same* sign for the fundamental and all slot/teeth/cage orders; reject infeasible sectors. Reference implementations ship with the paper sources.

Uniform-sign feasibility, 12-slot/4-pole, $\theta_s = 90^\circ$: signs conflict (−1 vs +1) -> infea

order m	physical meaning	sign $e^{-jm\pi/2}$	single sign serves all?
2	fundamental (working harmonic)	−1	yes
12	slot harmonic (k=1, odd)	+1	yes
24	slot harmonic (k=2, even)	−1	yes
36	slot harmonic (k=3, odd)	+1	yes

Figure 3: Uniform-sign feasibility table: single Link sign vs. significant mode order for the 12-slot/4-pole family. [Data: data/raw/case2_sector.]

9. What Is Obvious and What Is Not

That periodic boundaries should match harmonic periods is folklore, and simulator guides warn about it. What is not folklore, and what motivates this paper: (i) the failure is *silent* — every standard check passes; (ii) the error is *large and signed* (+44% on the fundamental); (iii) in coupled cage models the infeasibility can be *structural* (single Link sign cannot serve fundamental and teeth harmonics simultaneously): for the 12-slot/4-pole family the 90° sector is inadmissible under the uniform-sign condition, whereas 180° passes — the condition delineates admissible sector angles rather than banning them, and our executable feasibility check (`asm_spec.py`) encodes exactly this test; (iv) an analogous silence holds for F1 and F2, and no published protocol covers the three jointly for electrical-machine FE.

10. Implications for Simulation and Geometry Pipelines

(G1) bar-ratio normalization; (G2) sector pre-check; (G3) scale-band self-consistency; (G4) solver escalation before model changes. All four are implemented in our toolchain and referenced from the case forensics above.

11. Conclusion

Application-level silent failures in electrical-machine FE form a coherent class with a shared signature (no alarms) and complementary audits (cross-

solver, scale band, uniform sign). The three forensic cases, the executable protocol, and the archived data are released to make such audits routine.

Data availability

Raw solve logs, archived reports, generated figures, and reference implementations are bundled with the manuscript sources (`data/raw/`, `figure/`, `src/`).

Declaration of competing interest

The authors declare no competing interests.

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